(TR-104) PYTHON WITH AI/ML

Training Day 64 Report

Date: 9 April 2026

Topic Covered: Embedding System, Document Processing, Vector Database & Multi-LLM Workflow Enhancements

The sixty-fourth day of training focused on implementing an end-to-end embedding-based document processing pipeline, integrating vector storage using ChromaDB, and enhancing multi-LLM workflow support including Gemini integration. The session emphasized building scalable retrieval systems and improving workflow-based AI architecture.

- Implemented SentenceTransformer-based embedding system using all-MiniLM-L6-v2 model.
- Added safe model loading with proper error handling and structured logging.
- Built embedding generation pipeline for batch text processing and conversion into vector format.
- Handled edge cases such as empty input and model loading failures.
- Developed a structured Document class for storing text along with metadata.
- Implemented PDF processing using pdfplumber for accurate text extraction.
- Added table extraction support and converted tabular data into readable text format.
- Built noise removal logic to filter headers, footers, and page numbers from documents.
- Implemented HTML parsing using BeautifulSoup with fallback mechanisms for malformed content.
- Added plain text file loader with validation and preprocessing checks.
- Integrated ChromaDB as a persistent vector database for storing document embeddings.
- Implemented metadata-aware document indexing with chunk-level ID generation.

- Built a Chroma-based retrieval system using similarity search over embedded documents.
- Structured retrieval output with both content and associated metadata.
- Optimized embedding reuse using singleton EmbeddingManager for efficiency.
- Integrated Gemini model into multi-provider LLM routing system.
- Added Gemini API support within workflow engine for response generation.
- Implemented workflow save and load functionality for reusable pipeline configurations.
- Introduced Prompt Template Node for structured prompt-based generation.
- Tested Gemini integration with workflow execution successfully.
- Reviewed poster content for accuracy and suggested minor improvements in labeling and naming consistency.
- Strengthened understanding of RAG pipeline, embeddings, vector databases, and multi-LLM system integration.
- **GitHub Link:** https://github.com/JaspinderKaurWalia26/workflows
- **Blog Link:** https://medium.com/@jaspinderkaurwalia855/how-i-built-my-own-ai-workflow-system-using-a-knowledge-base-llm-tools-ec81c8afc77c